

DOUGLAS GUILHERME

Bioinformatics Machine Learning Expert

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Brazil

PROFESSIONAL SUMMARY

Bioinformatics Machine Learning Specialist with 5+ years building production-grade analytical-predictive models integrating artificial intelligence and genomic data for precision medicine and biopharmaceutical discovery. Owns bioinformatics pipelines end-to-end—from raw sequencing data processing and feature engineering to model deployment and clinical interpretation—while partnering with research scientists, clinicians, and cross-functional stakeholders. Ships scalable analytical platforms quickly, iterates from biological validation metrics, and maintains high-availability genomic data infrastructure with Docker, cloud computing, and robust observability tooling.

SKILLS

Machine Learning & AI	Deep Learning, Neural Networks, Clustering (k-means, DBSCAN, Mean Shift), Predictive Modeling, Feature Engineering, Hyperparameter Optimization, Dimensionality Reduction, Statistical Analysis	Bioinformatics	RNA-Seq Analysis, Variant Calling, Genome Assembly, Metagenomics, Transcriptomics, Proteogenomics, Molecular Docking, Virtual Screening, Microbiome Profiling
Programming	Python, R, SQL, Bash/Shell, JavaScript, Nextflow Workflow Management	Tools & Platforms	Google Cloud Platform (GCP), AWS, Docker, Kubernetes, Git, Jupyter, TensorFlow, PyTorch, Scikit-learn, Bioconductor, QIIME2, Prometheus, Grafana
Domains	Precision Medicine, Oncology Diagnostics, Drug Discovery, Infectious		

Disease Genomics, Public
Health Genomics,
Regulatory Compliance
(LGPD, HIPAA)

WORK EXPERIENCE

Bioinformatics Machine Learning Specialist

Jan 2024 – Present

Neoprospecta · Florianópolis, Brazil

- Led end-to-end delivery of machine learning pipelines for digital microbiological diagnostics, processing high-throughput sequencing data and coordinating requirements with quality control, R&D, and external clients across 3 release cycles.
- Architected and shipped predictive models using neural networks to detect contamination risks in production environments, implementing versioned model registries and validation frameworks that reduced false-positive rates by 32%.
- Built responsive analytical dashboards from wireframes while proposing UX refinements that cut diagnostic interpretation time by 40%.
- Designed PostgreSQL schemas and SQL migrations for genomic metadata and microbial reference databases, enabling scalable storage of 500,000+ sequencing samples with sub-second query performance.
- Containerized services with Docker and Kubernetes, establishing deployment pipelines that sustained 99.9% uptime across production and staging environments.
- Implemented event-driven workflows using Nextflow for genomic analysis, reducing end-to-end processing latency by 40% versus traditional batch processing.
- Established Prometheus and Grafana dashboards with SLI alerts on pipeline failure rates and processing latency, catching regressions before client impact in 85% of incidents.
- Mentored 2 junior bioinformaticians on coding standards and code review practices, raising automated test coverage on core analysis modules from 55% to 78%.

AI/ML Research Associate – Transcriptomics

Oct 2023 – Dec 2023

BioDecision Analytics · São Paulo, Brazil

- Delivered full stack analytical features for BDASeq platform, owning predictive models and cloud infrastructure from prototype through production deployment for transcriptomic data analysis.
- Integrated Google Cloud Platform services for scalable RNA-Seq data processing, designing error handling and idempotent data ingestion that eliminated duplicate analysis runs.
- Optimized database queries and indexes on BigQuery, reducing differential expression analysis report generation times from 12s to 2.3s for high-volume transcriptomic datasets.
- Partnered with research scientists and clinical collaborators in a remote, async workflow to scope MVP milestones, shipping biweekly releases with minimal post-release defects.

- Authored technical documentation and API examples for research partners, shortening time-to-first-analysis for new clients by 40%.
- Introduced comprehensive test coverage on core analysis modules, cutting regression bugs in production releases by 45% over 4 months.

Genomic Data Scientist

Jan 2022 – Sep 2023

DNA GTx Bioinformatics · Natal, Brazil

- Developed React and Python components for D-Krypt variant prioritization platform, improving variant interpretation efficiency by 35% through optimized feature extraction and memoized computation.
- Translated clinical requirements into accessible genomic analysis workflows, achieving 95% user satisfaction on primary diagnostic workflows reviewed by clinicians.
- Collaborated with backend engineers to define REST API contract shapes for variant annotation services, reducing frontend rework during API rollout by half.
- Built a reusable bioinformatics component library adopted across 3 product squads, standardizing variant calling and annotation patterns.
- Participated in sprint planning and stakeholder demos, delivering 20+ production features in a 20-month tenure with consistent on-time delivery.

Machine Learning Engineer – Structural Genomics

Aug 2020 – Dec 2021

Structural Genomics Laboratory (GenEstrut) – UFPel · Pelotas, Brazil

- Developed ML models for structural and functional genomics analysis, processing genomic and transcriptomic data for precision medicine and environmental genomics research applications.
- Built bioinformatics pipelines for microRNA biology and epigenetic data analysis, supporting patent development for molecular-based applications licensed to biotechnology companies.
- Contributed to Molecular Diagnostic Unit during COVID-19 pandemic, processing 10,000+ molecular tests and supporting regional public health response.
- Collaborated with 5+ graduate researchers to implement scalable genomic data processing workflows, reducing analysis bottlenecks and accelerating research outcomes.

EDUCATION

Master of Science in Bioinformatics

Universidade Federal do Rio Grande do Norte (UFRN) · Natal, Brazil · *Expected 2026*

Research focus: Integration of artificial intelligence and transcriptomic data for molecular target identification.

Bachelor's Degree in Biotechnology

Universidade Federal de Pelotas (UFPel) · Pelotas, Brazil · 2017 – 2021

Graduated with honors. Emphasis on structural genomics and computational biology.

SELECTED PUBLICATIONS & RESEARCH

AI-Enhanced Transcriptomic Discovery of Druggable Targets for Huntington's Disease – *Brain Sciences*, 2025 (Contributing Author)

BDASeq: A Scalable Analytical-Predictive Model for RNA-Seq Data Analysis – Presented at MENA Rare Disease Congress, UAE, 2024

Machine Learning Approaches in Precision Oncology Diagnostics – Manuscript under review, *Journal of Computational Biology*

CERTIFICATIONS

Machine Learning Specialization – Stanford University / Coursera

Deep Learning Specialization – deeplearning.ai

Genomic Data Science Specialization – Johns Hopkins University / Coursera

Nextflow & Scalable Bioinformatics Workflows – Seqera Labs